

Daily Sequences Drill #2 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Binomial & Trinomial Expansion · Pascal's Triangle as a Sequence · The Generalized Binomial Series.

Section A Rapid Recognition

1. **Answer:** 1, 5, 10, 10, 5, 1

Method: Pascal's rule: each entry is the sum of the two above it. Row 4 is 1, 4, 6, 4, 1; row 5 adds them pairwise with 1's at the ends.

Investigate further: Row n always has $n + 1$ entries and is symmetric: $\binom{n}{k} = \binom{n}{n-k}$.

2. **Answer:** 15

Method: $\binom{6}{2} = \frac{6 \times 5}{2!} = 15$.

Investigate further: Equivalently $\binom{6}{2} = \binom{6}{4}$ by symmetry — always check whether the “short side” of the ratio is faster to compute.

3. **Answer:** 2^n

Method: Setting $x = 1$ in $(1 + x)^n = \sum_k \binom{n}{k} x^k$ gives $2^n = \sum_k \binom{n}{k}$.

Investigate further: This is the single most useful substitution trick in the whole sheet — see the Top 5 Patterns below.

4. **Answer:** $1 + 3x + 3x^2 + x^3$

Method: Row 3 of Pascal's triangle is 1, 3, 3, 1.

Investigate further: Notice the coefficients are literally read off Pascal's triangle — no need to multiply out $(1 + x)(1 + x)(1 + x)$ by hand.

5. **Answer:** 15

Method: $\binom{6}{2} = 15$ (row 6: 1, 6, 15, 20, 15, 6, 1).

Investigate further: Same computation as A2 — recognising when two questions are secretly the same arithmetic saves real time.

6. **Answer:** No.

Method: Consecutive differences are $4 - 1 = 3$, $6 - 4 = 2$, $4 - 6 = -2$, $1 - 4 = -3$ — not constant, so the sequence is not arithmetic.

Investigate further: Pascal's triangle rows are symmetric and unimodal (rise then fall), which already rules out being arithmetic for any row with more than 2 entries and not all equal.

7. **Answer:** $1 + n$

Method: $\binom{n}{0} = 1$, $\binom{n}{1} = n$, so the sum is $1 + n$.

Investigate further: This is just the first two terms of the row-sum 2^n — compare to B2, which asks for the first four terms instead.

8. **Answer:** 0

Method: Setting $x = -1$ in $(1 + x)^n$: $(1 - 1)^n = 0^n = 0$ for $n \geq 1$, and this equals $\sum_k (-1)^k \binom{n}{k}$.

Investigate further: This is the “twin” substitution to A3 — $x = 1$ gives the row sum, $x = -1$ gives the alternating sum. Both are worth having on instant recall.

9. **Answer:** 1

Method: The constant term of $(1 + x)^5$ is the x^0 term, $\binom{5}{0} = 1$.

Investigate further: Every binomial expansion of $(1 + x)^n$ has constant term exactly 1, since $\binom{n}{0} = 1$ always.

10. **Answer:** True

Method: Row n has entries $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ — that's $n + 1$ entries, from $k = 0$ to $k = n$ inclusive.

Investigate further: A common off-by-one slip is saying row n has n entries — always count inclusively from 0.

Section B Manipulation Drills

1. **Answer:** Proof by substitution $x = 1$.

Method: $(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k$. Setting $x = 1$: $(1 + 1)^n = 2^n = \sum_{k=0}^n \binom{n}{k} (1)^k = \sum_{k=0}^n \binom{n}{k}$. $\square$

Investigate further: This is the cleanest possible proof of the identity — no induction needed, just one well-chosen substitution.

2. **Answer:** Neither, in general; the value is 176.

Method: $\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} = 1 + 10 + 45 + 120 = 176$. The four terms themselves (1, 10, 45, 120) have differences 9, 35, 75 (not constant) and ratios 10, 4.5, $2.\bar{6}$ (not constant), so the partial sum follows no simple pattern.

Investigate further: Contrast with A6: individual binomial coefficients within a row are neither arithmetic nor geometric — only the *full* row sum (2^n) and the full alternating sum (0) have clean closed forms.

3. **Answer:** A) 0

Method: $(1 + (-1))^n = 0^n = 0$ for $n \geq 1$.

Investigate further: At $n = 0$ the sum is just $\binom{0}{0} = 1$, not 0 — the identity needs $n \geq 1$ exactly because 0^0 is being avoided, not computed.

4. **Answer:** B) 40

Method: General term: $\binom{5}{k} 2^{5-k} x^k$. At $k = 3$: $\binom{5}{3} 2^2 = 10 \times 4 = 40$.

Investigate further: Don't forget the 2^{5-k} factor — the common slip is answering $\binom{5}{3} = 10$, treating the expansion as if it were $(1+x)^5$.

5. **Answer:** C) 240

Method: General term: $\binom{6}{k} (-2x)^k$. At $k = 4$: $\binom{6}{4} (-2)^4 = 15 \times 16 = 240$. The sign is positive since $(-2)^4$ is positive (even power).

Investigate further: The trap is B) -240 , from forgetting that an even power of a negative number is positive.

6. **Answer:** B) 6

Method: General term: $\binom{4}{k} x^{4-k} x^{-k} = \binom{4}{k} x^{4-2k}$. Constant term needs $4 - 2k = 0 \Rightarrow k = 2$: $\binom{4}{2} = 6$.

Investigate further: Whenever a binomial has both positive and negative powers of x , set the total exponent to zero and solve for k first — don't try to spot the constant term by expanding everything.

7. **Answer:** C) 43

Method: $\frac{\binom{n}{2}}{\binom{n}{1}} = \frac{n-1}{2} = 21 \Rightarrow n-1 = 42 \Rightarrow n = 43$.

Investigate further: This ratio trick — $\binom{n}{2}/\binom{n}{1} = (n-1)/2$ — is worth memorising directly rather than writing out both binomial coefficients in full every time.

8. **Answer:** B) 5

Method: $\binom{n}{2} = \binom{n}{3} \Rightarrow$ (by the symmetry $\binom{n}{k} = \binom{n}{n-k}$) 2 and 3 must be complementary: $n-2 = 3 \Rightarrow n = 5$. Check: $\binom{5}{2} = 10 = \binom{5}{3}$. ✓

Investigate further: This only works because $2 \neq 3$ forces the symmetry route — if the two given exponents were equal, any n would trivially work, and the question would be ill-posed.

9. **Answer:** C) 21

Method: Trinomial theorem: terms $(1)^a (2x)^b (3x^2)^c$ with $a + b + c = 3$, exponent $b + 2c = 2$. Case $(b, c) = (2, 0), a = 1$: coefficient $\frac{3!}{1!2!0!} 2^2 = 3 \times 4 = 12$. Case $(b, c) = (0, 1), a = 2$: coefficient $\frac{3!}{2!0!1!} 3^1 = 3 \times 3 = 9$. Total: $12 + 9 = 21$.

Investigate further: Every valid (a, b, c) triple with $a + b + c = 3$ and $b + 2c = 2$ must be found systematically — missing a case is the single most common trinomial-expansion error.

10. **Answer:** Geometric, common ratio 2.

Method: Row sums are $2^0, 2^1, 2^2, \dots = 1, 2, 4, 8, 16, \dots$ — each term is exactly double the previous one.

Investigate further: This directly answers A3 in sequence-language: the row sums themselves form a geometric sequence, even though no single row's entries do.

Section C Substitution & Structure

1. **Answer:** B) There is no simple closed form for a general partial sum of binomial coefficients, unlike the full row sum.

Method: The full sum ($k = n + 1$ terms) collapses via $x = 1$ substitution to 2^n ; a partial sum has no equivalent single substitution trick, and in general must be computed term by term (as in B2's 176).

Investigate further: The one genuine exception worth knowing: summing binomial coefficients down a *column* across many rows (rather than across one row) does have a closed form — the hockey-stick identity, which C3 explores.

2. **Answer:** C) 820

Method: The coefficient of x in $(1 + x)^k$ is $\binom{k}{1} = k$. Summing over $k = 0$ to 40: $\sum_{k=0}^{40} k = \frac{40 \times 41}{2} = 820$.

Investigate further: The real TMUA question (2019 Paper 1 Q3) sums to $k = 80$ instead of 40, giving $\frac{80 \times 81}{2} = 3240$. This is exactly Day 1's arithmetic series sum formula, now extracting information from a *sum of binomial expansions* rather than a plain AP — the two toolkits meet here.

3. **Answer:** B) $\binom{n+1}{3}$

Method: The coefficient of x^2 in $(1 + x)^k$ is $\binom{k}{2}$ (zero for $k < 2$). Summing $\sum_{k=2}^n \binom{k}{2} = \binom{n+1}{3}$ is the hockey-stick identity: summing a column of Pascal's triangle from its start lands one row down and one column right.

Investigate further: Try proving the hockey-stick identity itself by induction on n , using Pascal's rule $\binom{k}{2} + \binom{k+1}{2} \dots$ — or revisit C2, which is the same trick one column to the left ($\binom{k}{1} = k$ instead of $\binom{k}{2}$).

4. **Answer:** B) 16

Method: Trinomial theorem on $(1 + x + x^2)^4$: terms $(1)^a(x)^b(x^2)^c$, $a + b + c = 4$, exponent $b + 2c = 3$. Case $(b, c) = (3, 0)$, $a = 1$: $\frac{4!}{1!3!0!} = 4$. Case $(b, c) = (1, 1)$, $a = 2$: $\frac{4!}{2!1!1!} = 12$. Total: $4 + 12 = 16$.

Investigate further: Same systematic case-finding as B9, now with a larger power — always list every (b, c) pair satisfying the exponent equation before computing any coefficient.

5. **Answer:** B) $1 + x + x^2 + x^3 + \dots$

Method: Long division, or the generalized binomial series $(1 - x)^{-1} = \sum_{k=0}^{\infty} \binom{-1}{k} x^k$ with $(-x)^k$ sign conventions, both give every coefficient equal to 1.

Investigate further: This is the geometric series with $a = 1, r = x$ in disguise — the entire next sheet is built on recognising this connection. Check: multiplying $(1 - x)(1 + x + x^2 + \dots)$ should telescope to exactly 1.

6. **Answer:** B) 4

Method: At $k = 3$: coefficient is $k + 1 = 4$.

Investigate further: The coefficients $1, 2, 3, 4, \dots$ (i.e. $k + 1$) are themselves an arithmetic sequence in k with common difference 1 — differentiating $(1 - x)^{-1}$ once (which is exactly how $(1 - x)^{-2}$ arises) turns a constant coefficient sequence into a linear one.

7. **Answer:** D) $\sum_{k=0}^n k \binom{n}{k} = 2^n$

Method: The true identity is $\sum_{k=0}^n k \binom{n}{k} = n \cdot 2^{n-1}$ (proved in D3), not 2^n . Check $n = 3$: true value is $3 \times 4 = 12$, but D claims $2^3 = 8 \neq 12$.

Investigate further: A) is the row-sum identity (A3), B) is the alternating-sum identity (A8), C) is the symmetry identity (A1) — three genuine facts sitting beside one fabricated-looking but plausible fourth. Always test a small case rather than trusting a familiar-looking formula.

8. **Answer:** C) $\binom{7}{2}$

Method: $(1 - x)^{-3} = \sum_{k=0}^{\infty} \binom{k+2}{2} x^k$. At $k = 5$: $\binom{5+2}{2} = \binom{7}{2} = 21$.

Investigate further: Compare the exponents across $(1 - x)^{-1}, (1 - x)^{-2}, (1 - x)^{-3}$: the coefficient of x^k is $\binom{k}{0}, \binom{k+1}{1}, \binom{k+2}{2}$ respectively — a clean pattern worth remembering as “stars and bars in series form.”

Section D Challenge

1. **Answer:** A) 0

Method: General term of $(1 + x + y^2)^5$: $(1)^a (x)^b (y^2)^c$ with $a + b + c = 5$, contributing $x^b y^{2c}$. The power of y is always $2c$ — an even number. Since 3 is odd, there is no valid (a, b, c) giving $x^2 y^3$, so its coefficient is 0.

Investigate further: This is the real trap the real TMUA question (2020 Paper 1 Q13, which instead asks for the achievable $x^2 y^4$) is built to reward: recognising an *impossible* target power instantly, without attempting any case-by-case expansion, is faster and more reliable than grinding through the trinomial theorem only to find every case empty.

2. **Answer:** A) $a = \pm\sqrt{3}$

Method: Coefficient of x^2 in $(1 + x + 2x^2)^3$: cases $(b, c) = (2, 0), a=1$ gives $\frac{3!}{1!2!0!} 1^2 = 3$; $(b, c) = (0, 1), a=2$ gives $\frac{3!}{2!0!1!} 2^1 = 6$. Total $3 + 6 = 9$. Coefficient of x^2 in $(1 - ax)^3$: $\binom{3}{2} (-a)^2 = 3a^2$. Setting equal: $3a^2 = 9 \Rightarrow a^2 = 3 \Rightarrow a = \pm\sqrt{3}$.

Investigate further: The real TMUA question (Practice Paper 1 Q19) matches the coefficient of x^3 in $(1 + 2x + 3x^2)^6$ against *twice* the coefficient of x^4 in $(1 - ax^2)^5$, giving $a = \pm\sqrt{17}$ — same technique, larger trinomial power and an extra factor of 2. Work through that version by hand.

3. **Answer:** Proof by differentiating $(1 + x)^n$.

Method: $(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k$. Differentiating both sides with respect to x : $n(1 + x)^{n-1} =$

$\sum_{k=0}^n k \binom{n}{k} x^{k-1}$ (the $k=0$ term vanishes since it has no x). Setting $x=1$: $n \cdot 2^{n-1} = \sum_{k=0}^n k \binom{n}{k}$. $\square$

Investigate further: This is the standard trick for “weighted” binomial sums — differentiate first, then substitute. Try the same approach on $\sum_{k=0}^n k^2 \binom{n}{k}$ by differentiating twice (or by differentiating $x \cdot n(1+x)^{n-1}$ once more).

4. Answer: A) Vandermonde’s identity.

Method: $(1+x)^n(1+x)^m = (\sum_i \binom{n}{i} x^i) (\sum_j \binom{m}{j} x^j)$. The coefficient of x^k on the right is $\sum_{i=0}^k \binom{n}{i} \binom{m}{k-i}$ (pairing every way $i+j=k$). Since the left side is also $(1+x)^{n+m}$, whose coefficient of x^k is $\binom{n+m}{k}$, the two must be equal.

Investigate further: Vandermonde’s identity has a clean combinatorial reading too: choosing k people from a group of n men and m women, split by how many men are chosen — but this derivation gets there purely from series manipulation, no counting argument needed.

5. Answer: C) Quadratic in k (neither arithmetic nor geometric).

Method: $\binom{k+2}{2} = \frac{(k+2)(k+1)}{2} = \frac{1}{2}k^2 + \frac{3}{2}k + 1$, a genuine quadratic in k (nonzero k^2 coefficient).

Investigate further: This is exactly Day 1’s “ S_n is quadratic in n ” pattern (Day 1, C7/C8) reappearing in a completely different context: the coefficients of $(1-x)^{-1}, (1-x)^{-2}, (1-x)^{-3}$ are degree 0, 1, 2 polynomials in k respectively — each higher inverse power of $(1-x)$ raises the degree by exactly one.

Top 5 Patterns — Worksheet #2

- Row sum:** $\sum_k \binom{n}{k} = 2^n$, from setting $x=1$ in $(1+x)^n$. The single most useful substitution in the whole sheet. (see A3, B1, C2)
- Alternating sum:** $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \geq 1$, from setting $x=-1$. The “twin” of the row-sum trick. (see A8, B3)
- The generalized binomial series bridges straight into geometric series.** $(1-x)^{-m} = \sum_{k=0}^{\infty} \binom{k+m-1}{m-1} x^k$ — at $m=1$ this is exactly the geometric series. (see C5, C6, C8, D5)
- Trinomial/multinomial coefficient extraction: find every (a, b, c) with $a+b+c = \text{power}$ and the right exponent.** Missing a case is the single most common error. (see B9, C4, D1, D2)
- Differentiate $(1+x)^n$ term-by-term to extract weighted sums like $\sum k \binom{n}{k}$.** Substitute $x=1$ after differentiating, not before. (see C7, D3)

Common Traps — Worksheet #2

- Confusing a partial binomial sum with the full row sum.** Only the complete sum $k=0$ to n has the clean 2^n closed form — a partial sum in general has no shortcut. (see C1)
- Assuming every target power is achievable in a trinomial expansion.** $(1+x+y^2)^n$ can only ever produce *even* powers of y — an odd target power isn’t a computation to grind through, it’s genuinely zero. (see D1)
- Off-by-one errors between $\binom{n+1}{3}$ and $\binom{n}{3}$ in hockey-stick-style partial sums.** Summing a column of Pascal’s triangle lands one row *down*, not on the same row. (see C3)

4. **Assuming binomial-series coefficients stay linear (arithmetic) in k for every power of $(1-x)^{-m}$.** Only $m = 2$ gives an AP — higher m gives higher-degree polynomials in k . (see D5)
5. **Forgetting the multinomial coefficient needs all three factorials, $\frac{n!}{p!q!r!}$, not just a single $\binom{n}{k}$.** A trinomial expansion is not two nested binomial expansions in disguise. (see B9, C4)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths